```
-- ====================================================================
-- ASSIGNMENT III: SQL TABLE MANIPULATION
-- Project : Hotel Reservation System
-- DBMS    : PostgreSQL (pgAdmin 4)
-- Student : <YOUR FULL NAME> (202510491)
-- ====================================================================


-- ====================================================================
-- SECTION 1: DATABASE CREATION
-- ====================================================================
-- [DB1] Create the database
-- Objective: Create the database using the format yourname_studentid_projectname.
-- Run this on the default "postgres" database, then open a Query Tool on the new
one.
CREATE DATABASE samuel_202510491_hotel_reservation_system_database;


-- ====================================================================
-- SECTION 2: TABLE CREATION AND CONSTRAINTS (parents first, child last)
-- ====================================================================
-- [T1] Create the guests table (parent)
-- Objective: Store hotel guests. guest_id is generated automatically.
CREATE TABLE guests (
    guest_id    INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    full_name   VARCHAR(100) NOT NULL,
    phone       VARCHAR(15)  UNIQUE,
    nationality VARCHAR(50)
);

-- [T2] Create the rooms table (parent)
-- Objective: Store hotel rooms with their type and nightly price.
CREATE TABLE rooms (
    room_id         INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    room_number     VARCHAR(10) NOT NULL UNIQUE,
    room_type       VARCHAR(20) NOT NULL CHECK (room_type IN ('Single', 'Double',
'Deluxe')),
    price_per_night INT         NOT NULL CHECK (price_per_night > 0)
);

-- [T3] Create the reservations table (child)
-- Objective: Link guests to rooms. Two foreign keys: guest_id and room_id.
CREATE TABLE reservations (
    reservation_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
    guest_id       INT         NOT NULL REFERENCES guests(guest_id),
    room_id        INT         NOT NULL REFERENCES rooms(room_id),
```

```
    check_in       DATE        NOT NULL,
    nights         INT         NOT NULL CHECK (nights > 0),
    status         VARCHAR(20) NOT NULL DEFAULT 'Confirmed'
                   CHECK (status IN ('Confirmed', 'Checked-in', 'Completed',
'Cancelled'))
);

-- >>> GENERATE THE ERD NOW (before any ALTER TABLE task) <<<
-- pgAdmin 4: right-click the database -> ERD For Database -> Download image.
-- Save it as erd.png.


-- ======================================================================
-- SECTION 3: INITIAL DATA INSERTION (parents first, no id values typed)
-- ======================================================================
-- [I1] Insert 5 guests
-- Objective: Add the first five guests. Guest 5 has no reservation yet (used by the
LEFT JOIN).
INSERT INTO guests (full_name, phone, nationality) VALUES
('Alice Uwase',          '0788100001', 'Rwandan'),
('John Smith',           '0788100002', 'British'),
('Grace Wanjiru',        '0788100003', 'Kenyan'),
('Patrick Okello',       '0788100004', 'Ugandan'),
('Marie Claire Ingabire','0788100005', 'Rwandan');

-- [I2] Insert 6 rooms
-- Objective: Add six rooms. Room 302 is never booked at first (used by the LEFT
JOIN).
INSERT INTO rooms (room_number, room_type, price_per_night) VALUES
('101', 'Single', 40000),
('102', 'Single', 40000),
('201', 'Double', 70000),
('202', 'Double', 70000),
('301', 'Deluxe', 120000),
('302', 'Deluxe', 120000);

-- [I3] Insert 5 reservations
-- Objective: Add reservations using guest_id and room_id values that exist in the
parent tables.
INSERT INTO reservations (guest_id, room_id, check_in, nights, status) VALUES
(1, 1, '2026-10-10', 3, 'Completed'),
(2, 3, '2026-10-15', 2, 'Completed'),
(3, 5, '2026-10-20', 4, 'Checked-in'),
(4, 2, '2026-11-01', 5, 'Confirmed'),
(1, 4, '2026-11-10', 2, 'Confirmed');

-- [V1] Verify that every table has at least 5 rows and ids were generated
-- Objective: Count the rows of each table and view the generated keys.
SELECT 'guests' AS table_name, COUNT(*) AS total_rows FROM guests
UNION ALL SELECT 'rooms', COUNT(*) FROM rooms
```

```
UNION ALL SELECT 'reservations', COUNT(*) FROM reservations;

SELECT * FROM guests;
SELECT * FROM rooms;
SELECT * FROM reservations;



-- ========================================================================
-- SECTION 4: SELECT TASKS
-- ========================================================================
-- [Q1.1] SELECT - Basic data retrieval
-- Objective: Display each guest's name, phone and nationality.
SELECT full_name, phone, nationality
FROM guests;

-- [Q1.2] SELECT - Conditional data retrieval
-- Objective: Display only the Deluxe rooms.
SELECT room_number, room_type, price_per_night
FROM rooms
WHERE room_type = 'Deluxe';

-- [Q1.3] SELECT - Sorted data retrieval
-- Objective: Display all rooms from the most expensive to the cheapest.
SELECT room_number, room_type, price_per_night
FROM rooms
ORDER BY price_per_night DESC, room_number ASC;



-- ========================================================================
-- SECTION 5: JOIN TASKS
-- ========================================================================
-- [Q2.1] INNER JOIN
-- Objective: Show each reservation with the name of the guest who made it.
SELECT r.reservation_id, g.full_name, r.check_in, r.nights, r.status
FROM reservations r
INNER JOIN guests g ON r.guest_id = g.guest_id;

-- [Q2.2] LEFT JOIN
-- Objective: Show ALL guests, including those who have never made a reservation.
SELECT g.guest_id, g.full_name, r.reservation_id, r.check_in
FROM guests g
LEFT JOIN reservations r ON g.guest_id = r.guest_id
ORDER BY g.guest_id;

-- [Q2.3] Multi-table JOIN
-- Objective: Report guest, room, dates and total cost (nights x price) for every
reservation.
SELECT g.full_name, ro.room_number, ro.room_type, r.check_in, r.nights,
       r.nights * ro.price_per_night AS total_cost
FROM reservations r
```

```
JOIN guests g  ON r.guest_id = g.guest_id
JOIN rooms  ro ON r.room_id  = ro.room_id
ORDER BY r.check_in;


-- ========================================================================
-- SECTION 6: UPDATE TASKS AND VERIFICATION
-- ========================================================================
-- [Q3.1] UPDATE - One record
-- Objective: Change the phone number of guest 2 (John Smith).
-- Before: identify the record
SELECT guest_id, full_name, phone FROM guests WHERE guest_id = 2;
UPDATE guests
SET phone = '0788999999'
WHERE guest_id = 2;
-- Verify
SELECT guest_id, full_name, phone FROM guests WHERE guest_id = 2;

-- [Q3.2] UPDATE - Multiple records
-- Objective: Increase the price of all Single rooms by 10000.
-- Before: identify the records
SELECT room_id, room_number, room_type, price_per_night FROM rooms WHERE room_type =
'Single';
UPDATE rooms
SET price_per_night = price_per_night + 10000
WHERE room_type = 'Single';
-- Verify
SELECT room_id, room_number, room_type, price_per_night FROM rooms WHERE room_type =
'Single';

-- [Q3.3] UPDATE - Conditional update
-- Objective: Business rule: 'Confirmed' reservations with check-in on or before 5
Nov 2026 become 'Checked-in'.
-- Before: identify the records
SELECT reservation_id, check_in, status FROM reservations
WHERE status = 'Confirmed' AND check_in <= '2026-11-05';
UPDATE reservations
SET status = 'Checked-in'
WHERE status = 'Confirmed' AND check_in <= '2026-11-05';
-- Verify
SELECT reservation_id, check_in, status FROM reservations ORDER BY reservation_id;


-- ========================================================================
-- SECTION 7: ALTER TABLE TASKS AND VERIFICATION
-- ========================================================================
-- [Q4.1] ALTER TABLE - Rename a column
-- Objective: Rename guests.nationality to country.
ALTER TABLE guests RENAME COLUMN nationality TO country;
-- Verify: new name exists and the data is still there
```

```
SELECT column_name, data_type FROM information_schema.columns
WHERE table_name = 'guests' ORDER BY ordinal_position;
SELECT full_name, country FROM guests;

-- [Q4.2] ALTER TABLE - Change a data type
-- Objective: Change rooms.price_per_night from INT to NUMERIC(10,2) to allow decimal
prices.
-- INT converts safely to NUMERIC, so no value is lost.
ALTER TABLE rooms ALTER COLUMN price_per_night TYPE NUMERIC(10,2);
-- Verify
SELECT column_name, data_type, numeric_precision, numeric_scale FROM
information_schema.columns
WHERE table_name = 'rooms' AND column_name = 'price_per_night';
SELECT room_number, price_per_night FROM rooms ORDER BY room_number;

-- [Q4.3] ALTER TABLE - Add a column
-- Objective: Add an email column to guests so the hotel can send booking
confirmations.
ALTER TABLE guests ADD COLUMN email VARCHAR(100);
-- Verify
SELECT column_name, data_type, character_maximum_length FROM
information_schema.columns
WHERE table_name = 'guests' ORDER BY ordinal_position;



-- ========================================================================
-- SECTION 8: ADDITIONAL INSERT TASKS AND VERIFICATION
-- ========================================================================
-- [Q5.1] INSERT - Single record
-- Objective: Add one new guest (uses the renamed and new columns).
INSERT INTO guests (full_name, phone, country, email)
VALUES ('David Mugabo', '0788100006', 'Rwandan', 'david.mugabo@example.com');
-- Verify
SELECT * FROM guests WHERE full_name = 'David Mugabo';

-- [Q5.2] INSERT - Multiple records
-- Objective: Add three new rooms in one statement.
INSERT INTO rooms (room_number, room_type, price_per_night) VALUES
('103', 'Single', 50000.00),
('203', 'Double', 75000.00),
('303', 'Deluxe', 130000.00);
-- Verify
SELECT * FROM rooms ORDER BY room_id;

-- [Q5.3] INSERT - Related record
-- Objective: Add a reservation for guest 5 in room 6 (both exist in the parent
tables).
-- Check the parents first
SELECT guest_id, full_name FROM guests WHERE guest_id = 5;
SELECT room_id, room_number FROM rooms WHERE room_id = 6;
```

```
INSERT INTO reservations (guest_id, room_id, check_in, nights)
VALUES (5, 6, '2026-11-20', 3);
-- Verify (status was filled by its DEFAULT 'Confirmed')
SELECT * FROM reservations WHERE guest_id = 5;

-- Optional constraint test (this MUST fail: guest 999 does not exist):
-- INSERT INTO reservations (guest_id, room_id, check_in, nights) VALUES (999, 1,
'2026-12-01', 2);
```